

KALACH, Russian FederationWMO: **342470**Lat: **50.417N**Lon: **41.050E**Elev: **92**StdP: **100.23**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-21.1	-27.8	0.3	-24.3	-24.4	0.4	-20.7	9.3	-0.6	8.1	-3.3	1.0	0	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	33.7	19.2	31.6	19.1	29.8	18.5	21.4	28.9	20.6	28.1	19.8	27.1	3.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.0	23.8	18.2	13.2	23.2	17.4	12.5	22.6	62.4	28.9	59.5	28.0	57.0	27.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	6.0	-28.4	35.9	4.2	2.1	-31.4	37.4	-33.9	38.6	-36.3	39.8	-39.4	41.3		
(5)			-28.6	23.3	4.2	1.8	-31.6	24.6	-34.0	25.6	-36.4	26.7	-39.4	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-6.3	-6.0	-0.2	8.9	15.6	19.5	21.7	20.4	14.3	7.4	1.1	-4.1		
	DBStd	11.34	6.94	7.20	5.05	4.51	4.47	3.87	3.24	3.81	4.18	4.58	5.37	6.46		
	HDD10.0	2171	504	449	317	71	7	0	0	0	11	106	267	438		
	HDD18.3	4156	762	682	575	282	108	33	7	23	133	339	516	697		
	CDD10.0	1356	0	0	1	39	181	284	363	322	138	26	2	0		
	CDD18.3	298	0	0	0	0	23	66	112	87	11	0	0	0		
	CDH23.3	3547	0	0	0	14	323	752	1209	1070	177	2	0	0		
	CDH26.7	1357	0	0	0	1	91	272	500	452	42	1	0	0		
(14) Wind	WSAvg	2.4	2.7	2.9	2.9	2.7	2.4	2.1	1.8	1.9	2.1	2.2	2.7	2.8		
(15) Precipitation	PrecAvg	502	45	36	31	29	38	54	58	38	49	41	40	44		
	PrecMax	690	90	69	67	111	61	116	145	93	127	91	78	88		
	PrecMin	348	14	9	5	4	16	14	10	2	3	5	2	13		
	PrecStd	93	21	16	15	21	14	27	32	23	37	23	23	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	7.1	15.0	25.2	31.6	34.2	36.6	37.5	30.3	22.3	14.1	8.1		
		MCWB	5.6	5.3	7.9	13.6	17.1	18.7	20.2	19.4	16.7	14.2	11.4	6.7		
	2%	DB	3.8	4.4	11.7	22.1	29.2	31.7	33.8	33.7	27.7	19.5	11.8	6.2		
		MCWB	2.8	3.1	6.7	12.6	16.7	19.0	19.8	19.0	15.6	13.1	10.0	5.3		
	5%	DB	2.5	2.9	8.8	19.7	27.0	29.8	31.4	31.3	25.3	17.2	9.9	4.5		
		MCWB	1.5	1.8	5.6	11.3	16.0	18.7	19.7	18.8	15.4	12.6	8.2	3.6		
	10%	DB	1.4	1.9	6.3	17.5	24.7	27.8	29.4	29.0	22.9	14.7	8.0	2.8		
		MCWB	0.7	1.0	3.8	10.5	15.6	18.3	19.1	18.3	14.8	11.3	6.7	2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	5.8	9.6	15.4	19.2	22.2	22.5	22.1	19.3	16.4	12.1	7.0		
		MCDB	6.7	7.1	13.2	22.2	26.7	29.2	30.1	29.9	26.0	20.1	13.4	7.5		
	2%	WB	3.1	3.2	7.3	13.6	18.2	21.0	21.5	20.7	17.9	14.6	10.2	5.4		
		MCDB	3.6	4.1	10.9	19.6	25.9	27.8	29.4	28.7	23.3	17.5	11.7	6.2		
	5%	WB	1.7	2.0	5.5	12.5	17.2	20.0	20.8	19.9	16.7	13.0	8.5	3.8		
		MCDB	2.3	2.7	8.2	18.1	24.6	26.7	28.4	28.3	21.7	16.0	9.7	4.5		
	10%	WB	0.8	1.1	4.0	11.2	16.2	19.0	20.1	19.1	15.5	11.5	6.8	2.1		
		MCDB	1.3	1.7	6.0	16.3	23.0	25.7	27.4	26.8	21.2	14.5	7.9	2.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.8	7.6	11.4	12.3	11.8	12.1	13.7	12.5	9.0	5.8	5.1		
		MCDBR	3.6	5.1	11.7	15.5	15.9	15.2	15.7	17.1	17.2	13.6	8.3	4.5		
	5% WB	MCWBR	3.5	4.4	7.4	7.9	6.2	5.7	5.4	5.5	7.5	8.0	6.3	4.1		
		MCWBR	3.5	4.5	10.2	12.9	13.1	12.3	12.8	14.2	12.5	11.0	7.0	4.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.352	0.379	0.446	0.410	0.400	0.428	0.438	0.396	0.368	0.345	0.318		
		taud	2.101	2.006	2.077	2.078	2.276	2.334	2.261	2.214	2.308	2.331	2.270	2.169		
	Ebn at Noon	Ebn at Noon	673	743	806	793	847	858	825	794	789	739	650	614		
		Edh at Noon	82	117	132	148	127	121	128	128	105	86	71	67		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.03	2.03	3.35	4.37	5.80	6.19	6.03	5.37	3.65	1.96	0.89	0.60		
		RadStd	0.21	0.36	0.51	0.42	0.50	0.49	0.42	0.39	0.45	0.29	0.18	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	-144	+106	N/A

Nomenclature: See separate page